

CLASS-8

LESSON-10 EXPONENTS AND POWERS

(This PDF Based on NCERT Book)

EXPONENTS, BASE AND POWERS(घातांक, आधार और घात)-

Exponent (or Index/Power): This is the small, raised number that tells you **how many times** to multiply the main number by itself.

- *Example:* In the expression 5^3 , the **3** is the exponent.

Base: This is the main number that is being multiplied repeatedly.

- *Example:* In the expression 5^3 , the **5** is the base.

Power: This term is often used interchangeably with "exponent," but it technically refers to the **entire expression** (the base and the exponent together) or the **result** of the calculation.

- *Example:* The entire expression 5^3 is a power (read as "5 to the power of 3"). The result of $5 \times 5 \times 5 = 125$ is also called the power.

POWERS WITH NEGATIVE EXPONENTS(ऋणात्मक घातांक वाली घाते)-

Powers with **negative exponents** are a simple way to represent **reciprocals** or **fractions**.

- **Reciprocal** means "1 divided by that number."

Formula:

$$a^{-n} = \frac{1}{a^n}$$

Example

Let's look at 5^{-2} .

1. **The Base is 5.**
2. **The Exponent is -2.** The negative sign tells you to flip the number (take the reciprocal).
3. **The Reciprocal:** Flip the base (5) and change the exponent to positive 2.

$$5^{-2} = \frac{1}{5^2}$$

Now, you solve the power in the denominator (the bottom of the fraction):

$$\frac{1}{5^2} = \frac{1}{5 \times 5} = \frac{1}{25}$$

So, 5^{-2} is just another way of writing $\frac{1}{25}$.

LAWS OF EXPONENTS(घातांकों का नियम)-

The Core Laws of Exponents

Law	In Simple Words	Example	Formula
Product Rule	When you multiply powers with the same base , you add the exponents.	$2^3 \times 2^4 = 2^{(3+4)} = 2^7$	$a^m \times a^n = a^{(m+n)}$
Quotient Rule	When you divide powers with the same base , you subtract the exponents.	$\frac{5^6}{5^2} = 5^{(6-2)} = 5^4$	$\frac{a^m}{a^n} = a^{(m-n)}$
Power Rule	When you raise a power to another power , you multiply the exponents.	$(3^2)^5 = 3^{(2 \times 5)} = 3^{10}$	$(a^m)^n = a^{mn}$

Special Exponent Rules

Rule	In Simple Words	Example	Formula
Zero Exponent	Any number (except 0) raised to the power of zero is always 1 .	$100^0 = 1$	$a^0 = 1$
Negative Exponent	A negative exponent means you take the reciprocal (flip it into a fraction) to make the exponent positive .	$4^{-3} = \frac{1}{4^3}$	$a^{-m} = \frac{1}{a^m}$
Product to a Power	When a group of multiplied numbers is raised to a power, the power applies to EACH number inside.	$(3x)^2 = 3^2 x^2 = 9x^2$	$(ab)^m = a^m b^m$
Quotient to a Power	When a fraction is raised to a power, the power applies to both the top (numerator) and the bottom (denominator).	$\left(\frac{2}{5}\right)^3 = \frac{2^3}{5^3} = \frac{8}{125}$	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

USE OF EXPONENTS TO EXPRESS SMALL NUMBERS IN STANDARD FORM(मानक रूप में छोटी संख्याओं को व्यक्त करने के लिए घातांकों का उपयोग)-

Standard form always writes a number as a product of two parts:

First Part $\times 10^{\text{Exponent}}$

1. **First Part (The Coefficient):** This must be a number between 1 and 10 (it can be 1, but must be less than 10).
2. **Exponent (The Power of 10):** This tells you how many places and in which direction to move the decimal point.

Exponents Express Small Numbers

For a very small number (like 0.00045), the key is using a **negative exponent**.

A **negative exponent** (like 10^{-4}) means you are **dividing** by 10 repeatedly. Since dividing by 10 makes a number smaller, this is the perfect way to represent a decimal number that is less than one.

The Steps in Simple Words:

Let's use the small number **0.0000037** as an example.

1. **Find the First Part:** Move the decimal point **to the right** until it is just after the **first non-zero digit** (the 3).
 - The new number is **3.7**. This is your first part.
2. **Count the Moves:** Count how many places you moved the decimal point.
 - In this case, you moved it 6 places: 0. 0 0 0 0 0 **3. 7**
3. **Use a Negative Exponent:** Since you moved the decimal **right** to make the number larger (3.7), you must use a **negative** exponent to show that the original number was small. The count (6) becomes the exponent.

Original Number: 0.0000037

Standard Form: 3.7×10^{-6}

The -6 tells you that if you multiply 3.7 by $\frac{1}{10^6}$ (or divide it by 1,000,000), you get the tiny original number back.

COMPARING VERY LARGE AND VERY SMALL NUMBER(बहुत बड़ी और बहुत छोटी संख्याओं की तुलना)-

Comparing Exponents(घातांकों के तुलना)

When numbers are written as $N \times 10^n$, you compare them by looking at the exponent, n , first.

1. The Exponent (The Power of 10)

The exponent is the most important part because it tells you the **order of magnitude**—it defines how many places the number is away from 1.

- **Rule:** The number with the **larger exponent** is the **greater number**, no matter what the first part (N) is.
 - This is true even if the exponents are negative!

Example	Comparison	Simple Reason
2×10^8 vs. 9×10^5	2×10^8 is greater	8 is greater than 5. The 10^8 number is in the <i>hundreds of millions</i> ; the 10^5 number is only in the <i>hundreds of thousands</i> .
5×10^{-4} vs. 1×10^{-6}	5×10^{-4} is greater	-4 is greater than -6. The 10^{-4} number is less negative , so it's closer to zero (a bigger number).

Comparing Coefficients(गुणकों की तुलना)

If the exponents (n) are exactly the same, they have the same scale. Then you only need to look at the **first part** of the number (N, which is always between 1 and 10).

2. The Coefficient (The N part)

- **Rule:** If the exponents are equal, the number with the **larger coefficient** is the **greater number**.

Example	Comparison	Simple Reason
7.5×10^{12} vs. 2.1×10^{12}	7.5×10^{12} is greater	Both are in the trillions (10^{12}), but 7.5 is greater than 2.1.
3.0×10^{-9} vs. 9.9×10^{-9}	9.9×10^{-9} is greater	Both are tiny numbers in the same scale, but 9.9 is greater than 3.0.